WORKED SOLUTION

STEP III 2017, Question 2

Pure • Rotations in the Complex Plane

STEP III 2017, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q2

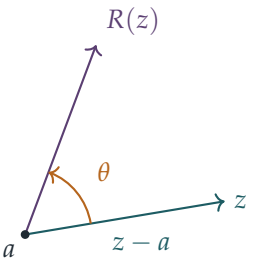
The key idea

A rotation about a is easiest to describe by moving the origin to a: subtract a, multiply by $e^{i\theta}$, then add a back. Every rotation therefore has the form $z \mapsto e^{i\alpha}z + k$, and conversely such a map is a rotation by α about its fixed point (provided $e^{i\alpha} \neq 1$). So to identify SR we compose the two formulae and find the fixed point. The half-angle factorisation $1 - e^{i\alpha} = -2i\, e^{i\alpha/2} \sin \frac{1}{2}\alpha$ turns the answer into the required shape.

Solution

Part (i)

The vector from the centre a to P is $z - a$. Rotating anticlockwise by θ about a rotates this vector by θ, which in complex numbers is multiplication by $e^{i\theta}$; the length is unchanged since $|e^{i\theta}| = 1$.



So the image w of P satisfies $w - a = e^{i\theta}(z - a)$, that is

$$w = e^{i\theta}z + a(1 - e^{i\theta}),$$

as required. In the same way $S(z) = e^{i\phi}z + b(1 - e^{i\phi})$.

Part (ii)

Apply R first and then S:

$$\begin{aligned} SR(z) &= \mathrm{e}^{\mathrm{i}\phi}\left(\mathrm{e}^{\mathrm{i}\theta}z + a(1-\mathrm{e}^{\mathrm{i}\theta})\right) + b(1-\mathrm{e}^{\mathrm{i}\phi}) \\ &= \mathrm{e}^{\mathrm{i}(\theta+\phi)}z + a\,\mathrm{e}^{\mathrm{i}\phi}(1-\mathrm{e}^{\mathrm{i}\theta}) + b(1-\mathrm{e}^{\mathrm{i}\phi}). \end{aligned}$$

By part (i), a rotation by $\theta+\phi$ about c is $z \mapsto \mathrm{e}^{\mathrm{i}(\theta+\phi)}z + c(1-\mathrm{e}^{\mathrm{i}(\theta+\phi)})$. The coefficients of z already agree, so SR is this rotation exactly when

$$c(1-\mathrm{e}^{\mathrm{i}(\theta+\phi)}) = a\,\mathrm{e}^{\mathrm{i}\phi}(1-\mathrm{e}^{\mathrm{i}\theta}) + b(1-\mathrm{e}^{\mathrm{i}\phi}). \qquad (*)$$

? **Ask yourself** How can we turn the brackets $1-\mathrm{e}^{\mathrm{i}\alpha}$ into sines of half-angles?

Take out a factor $\mathrm{e}^{\mathrm{i}\alpha/2}$ to make the bracket symmetric:

$$1-\mathrm{e}^{\mathrm{i}\alpha} = \mathrm{e}^{\mathrm{i}\alpha/2}(\mathrm{e}^{-\mathrm{i}\alpha/2} - \mathrm{e}^{\mathrm{i}\alpha/2}) = -2\mathrm{i}\,\mathrm{e}^{\mathrm{i}\alpha/2}\sin\tfrac{1}{2}\alpha.$$

Using this with $\alpha = \theta+\phi$, θ and ϕ, equation $(*)$ becomes

$$-2\mathrm{i}\,c\,\mathrm{e}^{\mathrm{i}(\theta+\phi)/2}\sin\tfrac{1}{2}(\theta+\phi) = -2\mathrm{i}\,a\,\mathrm{e}^{\mathrm{i}\phi}\mathrm{e}^{\mathrm{i}\theta/2}\sin\tfrac{1}{2}\theta - 2\mathrm{i}\,b\,\mathrm{e}^{\mathrm{i}\phi/2}\sin\tfrac{1}{2}\phi.$$

Dividing through by $-2\mathrm{i}\,\mathrm{e}^{\mathrm{i}(\theta+\phi)/2}$ (never zero) gives

$$c\sin\tfrac{1}{2}(\theta+\phi) = a\,\mathrm{e}^{\mathrm{i}\phi/2}\sin\tfrac{1}{2}\theta + b\,\mathrm{e}^{-\mathrm{i}\theta/2}\sin\tfrac{1}{2}\phi.$$

If $\theta+\phi \neq 2n\pi$ then $\sin\frac{1}{2}(\theta+\phi) \neq 0$, so this determines a unique c, and with that c the map SR is the rotation by $\theta+\phi$ about c. This is the required result. (The condition matters: if $\sin\frac{1}{2}(\theta+\phi) = 0$ there is no fixed point to rotate about, in general.)

The case $\theta+\phi = 2\pi$. Now $\mathrm{e}^{\mathrm{i}(\theta+\phi)} = 1$ and $\mathrm{e}^{\mathrm{i}\phi} = \mathrm{e}^{-\mathrm{i}\theta}$, so

$$\begin{aligned} SR(z) &= z + a\,\mathrm{e}^{-\mathrm{i}\theta}(1-\mathrm{e}^{\mathrm{i}\theta}) + b(1-\mathrm{e}^{-\mathrm{i}\theta}) \\ &= z + a(\mathrm{e}^{-\mathrm{i}\theta} - 1) + b(1-\mathrm{e}^{-\mathrm{i}\theta}) \\ &= z + (b-a)(1-\mathrm{e}^{-\mathrm{i}\theta}). \end{aligned}$$

So SR is a *translation* by $(b-a)(1-\mathrm{e}^{-\mathrm{i}\theta})$, a distance $2|b-a|\,|\sin\frac{1}{2}\theta|$. It is the identity when $a=b$ (rotating by θ and then by $2\pi-\theta$ about the same point) or when θ is a multiple of 2π; otherwise it is a genuine translation with no fixed point.

Part (iii)

Swapping the roles of (a,θ) and (b,ϕ) in the formula for SR,

$$RS(z) = \mathrm{e}^{\mathrm{i}(\theta+\phi)}z + b\,\mathrm{e}^{\mathrm{i}\theta}(1-\mathrm{e}^{\mathrm{i}\phi}) + a(1-\mathrm{e}^{\mathrm{i}\theta}).$$

Both maps multiply z by $\mathrm{e}^{\mathrm{i}(\theta+\phi)}$, so $RS = SR$ exactly when the constant terms agree. Their difference is

$$\begin{aligned} SR(z) - RS(z) &= a(1-\mathrm{e}^{\mathrm{i}\theta})(\mathrm{e}^{\mathrm{i}\phi}-1) + b(1-\mathrm{e}^{\mathrm{i}\phi})(1-\mathrm{e}^{\mathrm{i}\theta}) \\ &= (b-a)(1-\mathrm{e}^{\mathrm{i}\theta})(1-\mathrm{e}^{\mathrm{i}\phi}). \end{aligned}$$

A product is zero only if a factor is zero, so $RS = SR$ if and only if

- $a = b$ (the rotations have the same centre), or
- $\mathrm{e}^{\mathrm{i}\theta} = 1$, i.e. θ is a multiple of 2π, so R is the identity, or
- $\mathrm{e}^{\mathrm{i}\phi} = 1$, i.e. ϕ is a multiple of 2π, so S is the identity.

In words: two genuine rotations commute only when they share a centre.

Checks

- With $a = b$ the formula in (ii) gives $c \sin\frac{1}{2}(\theta+\phi) = a\big(\mathrm{e}^{\mathrm{i}\phi/2}\sin\frac{1}{2}\theta + \mathrm{e}^{-\mathrm{i}\theta/2}\sin\frac{1}{2}\phi\big)$, and expanding the bracket gives $\sin\frac{1}{2}\theta\cos\frac{1}{2}\phi + \cos\frac{1}{2}\theta\sin\frac{1}{2}\phi = \sin\frac{1}{2}(\theta+\phi)$ (the imaginary parts cancel), so $c = a$, as it must.
- With $\phi = 0$ (S the identity), $c\sin\frac{1}{2}\theta = a\sin\frac{1}{2}\theta$, so $c = a$.
- Take $a = 0$, $b = 1$, $\theta = \phi = \frac{1}{2}\pi$. Then $R(z) = \mathrm{i}z$, $S(z) = \mathrm{i}z + 1 - \mathrm{i}$, so $SR(z) = -z + 1 - \mathrm{i}$, a half-turn about $\frac{1}{2}(1-\mathrm{i})$. The formula gives $c \cdot 1 = \mathrm{e}^{-\mathrm{i}\pi/4}\cdot\frac{1}{\sqrt{2}} = \frac{1}{2}(1-\mathrm{i})$. Agreed.
- All three results were also confirmed by computer on thousands of random centres, angles and points, rotating geometrically (by modulus and argument) rather than with the formulae; the discrepancies were below 10^{-12}.

Common slips

- Composing in the wrong order: SR means R first, so $R(z)$ is substituted into S.
- Dividing by $1 - \mathrm{e}^{\mathrm{i}(\theta+\phi)}$ without noting that it is non-zero precisely because $\theta + \phi \neq 2n\pi$.
- In (iii), forgetting the trivial cases where one rotation is the identity.

Beyond STEP

The maps $z \mapsto \alpha z + \beta$ with $|\alpha| = 1$ are exactly the orientation-preserving isometries of the plane, and they form a group: rotations and translations together, closed under composition. Part (iii) is a small instance of the fact that this group is not commutative, and the commutator $RSR^{-1}S^{-1}$ turns out to be a translation. Needham's *Visual Complex Analysis* develops this geometric view of complex multiplication beautifully, and Allenby's *Rings, Fields and Groups* covers the group theory.

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